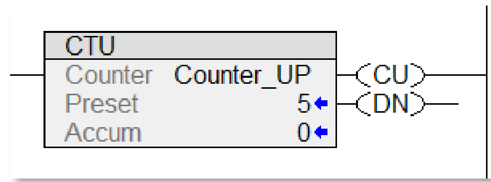
Counter Instructions Summery

Studio 5000 COUNTERS

Counter Up (CTU)

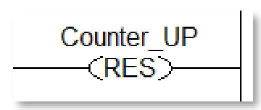


The Counter Up (**CTU**) is an output instruction that will turn **on** a controlled output when the accumulated value is greater than or equal to the preset value.

The greater than or equal to value will result in the **Done** bit being **set**. The Done bit may be used to control other logic.

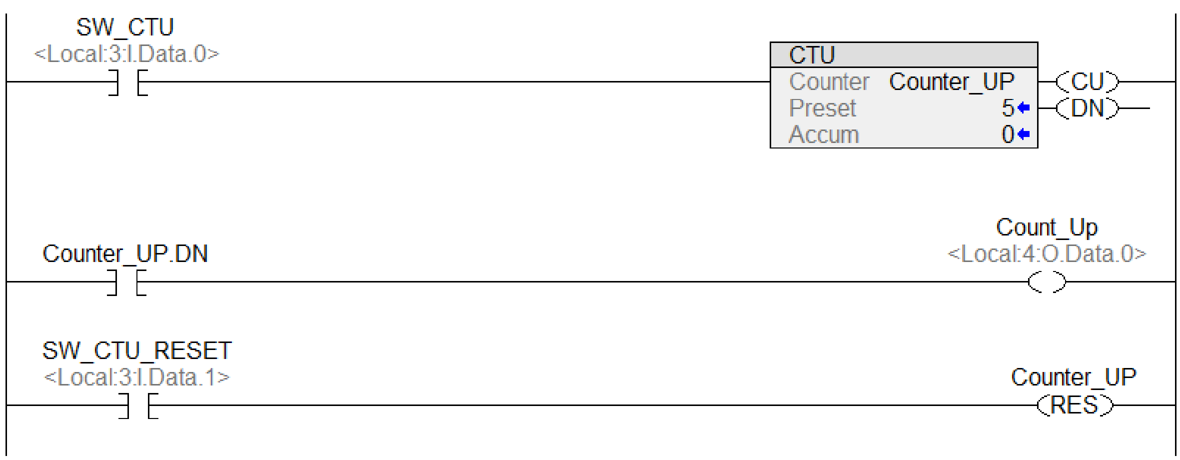
When the rung transitions from false to true, the counter accumulated value increments by one.

Counter Reset (RES)



When the rung is **TRUE**, the Reset (**RES**) instruction returns the Counter Up (**CTU**) with the same Tag Accumulated value to zero.

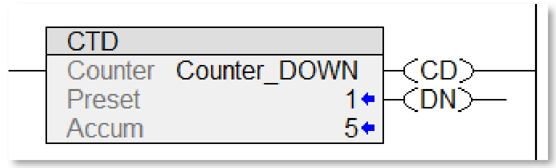
Counter Up Example



Status Bits

Counter Up (CTU)	
Counter Up Enable Bit **.CU**	**TRUE** when counter rung True
Done Bit **.DN**	**TRUE** when Accumulated (ACC) value is greater than or equal the Preset (PRE) value

Counter Down (CTD)



The Counter Down (**CTD**) is an output instruction that will turn **on** a controlled output while the accumulated value is greater than the preset value.

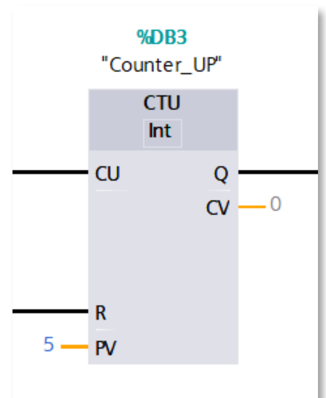
The accumulated value is greater or equal to the preset value will result in the **Done** bit being **set**. The Done bit may be used to control other logic.

When the rung transitions from false to true, the counter accumulated value decrements by one.

Up-Down Counter is created by combining an Up Counter with a Down Counter. Both counters use the same Counter Data Type Tag. The **.DN** status bit is controlled by the Up Counter. The counters are controlled by independent inputs.

Siemens COUNTERS

Counter Up (CTU)



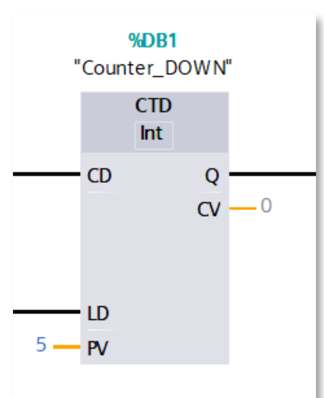
The Counter Up (**CTU**) is an instruction that will turn **on** a controlled output when the Count Value (**CV**) is greater than or equal to the Preset Value (**PV**).

The equal values will result in the Output Bit (**QU**) being **set**. The Output Bit is used to control other logic. Note: **Q** shown in instruction, use **QU** when assigning Tag.

The Reset (**R**) returns the Count Value to zero.

When the rung transitions from false to true, the counter increments.

Counter Down (CTD)



The Counter Down (**CTD**) is an instruction that will turn **on** a controlled output when the Count Value (**CV** is equal to or less than zero.

The equal values will result in the Output Bit (**QD**) being **set**. The Output Bit is used to control other logic. Note: **Q** shown in instruction, use **QD** when assigning Tag.

The Load (**LD**) returns the Count Value to the preset Value.

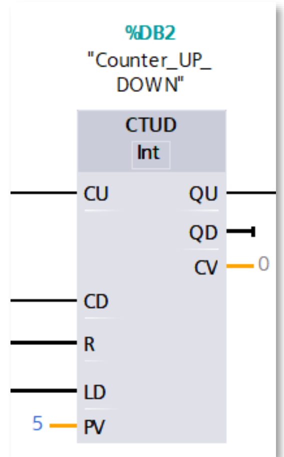
When the rung transitions from false to true, the counter decrements.

Counter Up/Down (CTUD)

The Counter Up/Down (**CTUD**) is an instruction that will turn **on** a controlled output (**QU**) when the Counter Value (**CV**) is greater than or equal to the Preset Value (**PV**).

A Counter Value (**CV**) less than or equal to the Preset Value (**PV**) will turn **on** a controlled output (**QD**).

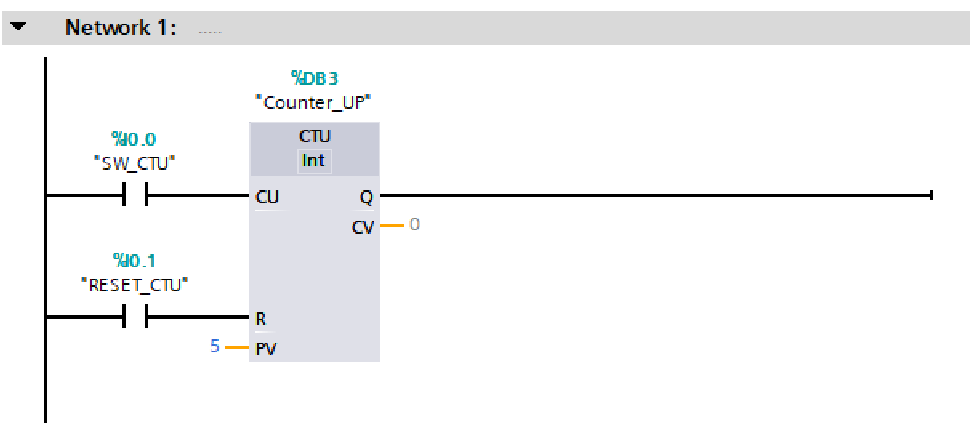
When the rung transitions from false to true, the counter increments or decrements, depending on the use of the Count Up (CU) or Count Down (CD).

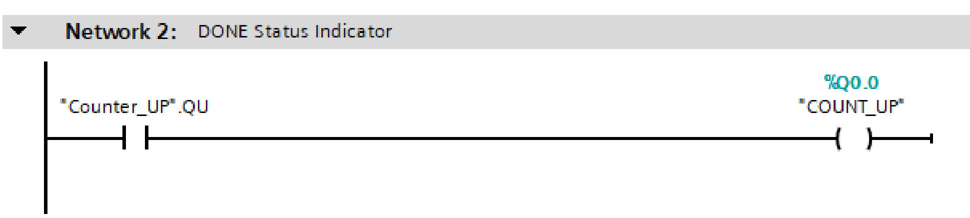


Status Bits

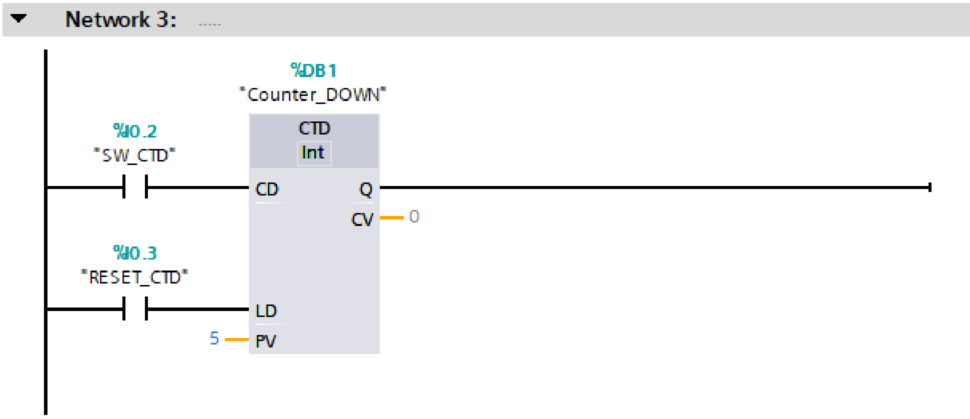
Counter Up (CTU)	
Counter Up Input Bit **.CU**	**TRUE** when counter rung True
Count Status Bit **.QU**	**TRUE** when the Count Value (CV) is greater than or equal to the Preset Value (PV).
Counter Down (CTD)	
Counter Down Input Bit **.CD**	**TRUE** when counter rung True
Count Status Bit **.QD**	**TRUE** when the Count Value (**CV** is equal to or less than zero

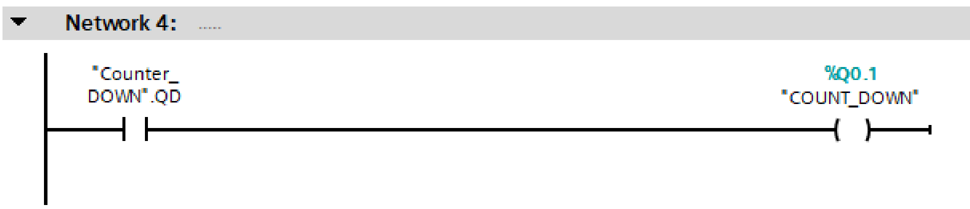
Counter Up Example





Counter Down Example





Counter Up/Down Example